OFFICIAL TRAINING PROPOSAL | HANDS-ON STEM KIT | AGES 8–12

ILLUSTRATED MANUAL & BONUS PROJECTS

Electronics STEM Kit for Kids (Ages 8–12)

Educational Aim: To make students familiar with fundamental concepts of electronics and circuits, preparing them with confidence for future robotics, automation, and advanced engineering courses.

📘 **Comprehensive User Manual Included:** Each kit includes a detailed, child-friendly illustrated guidebook featuring step-by-step wiring guides, safety rules, and instructions for **many additional DIY projects** that can be built at home using these exact components!

Offered By: CS Technologies

Trainers: Engr. Aamir Jamil (Lead AI, Robotics & All STEAM) & Miss Zia Ul Ain (Supportive Trainer)

Duration: 2 Lessons (2 Hours each = 4 Hours)

Power Source: Safe Low-Voltage AA Batteries

User Manual: Detailed step-by-step guide with diagrams

Bonus Projects: Expands into endless home experiments



6+	4	14+
Hands-On Projects	Learning Hours	Parts & Manual

✓ Zero Coding ✓ No Arduino ✓ No Soldering

ACCREDITED TRAINERS

Meet the Trainers

Expert educators from CS Technologies specializing in experiential STEAM, robotics, AI, and child-safe hardware prototyping.

LEAD TRAINER

Engr. Aamir Jamil

Lead AI, Robotics & STEAM Trainer

CS Technologies

Experienced, professional AI and robotics trainer and lead with comprehensive expertise across all STEAM disciplines. Dedicated to demystifying electronics, intelligent automation, and child-safe robotics for young learners through hands-on project creation and intuitive scientific experiments.

Lead AI & Robotics | All STEAM Disciplines | Circuit & Sensor Prototyping | Hands-on Engineering

SUPPORTIVE TRAINER

Miss Zia Ul Ain

Supportive Trainer

CS Technologies

Supportive trainer assisting during workshop sessions, providing hands-on student mentoring, circuitry troubleshooting, and guiding young learners through hardware assembly and safe experiment execution.

Supportive Trainer | Lab Assistance | Breadboard Guidance | Student Mentorship

CORE COMPETENCIES

What Students Will Learn

A proven progressive pathway covering circuit theory, active electronic components, sensors, and practical troubleshooting.

Component Mastery

Identify breadboards, LEDs, resistors, switches, buzzers, tilt sensors, and DC motors by sight and function.

Circuit Principles

Understand voltage, current, resistance, LED polarity (anode/cathode), and the role of current-limiting resistors.

Sensors & Motion

Harness light sensors (LDRs) and orientation tilt switches to trigger real-world alarms and spin motor fans.

Fault Troubleshooting

Form scientific curiosity by debugging loose connections, reversed polarity, open loops, and loose jumper leads.

CURRICULUM PORTFOLIO

6 Hands-on Engineering Projects

Step-by-step builds using child-safe components on a breadboard. Click any image to view in full resolution.



Time: 25 mins Output: Light

Mini LED Torch

Objective: Build a complete battery-powered handheld torch while mastering LED polarity and resistor protection.

COMPONENTS USED

- 2 × AA Batteries
- Battery Holder
- 1 × Red/White LED
- 1 × 330Ω Resistor
- Jumper Wires
- Cardboard Housing

BUILD PROCEDURE

1. Identify positive (+) and negative (-) battery terminals.
2. Determine LED anode (long leg) and cathode (short leg).
3. Wire battery positive to resistor, then to LED anode.
4. Connect LED cathode back to battery negative to close the circuit.
5. Mount securely into custom cardboard torch structure.

Learning: Complete circuits, LED polarity & resistor safety.



Time: 25 mins Output: Controlled Light

Push-Button Light

Objective: Control an LED dynamically using an interactive push-button switch to grasp open vs. closed circuits.

COMPONENTS USED

- Mini Breadboard
- Push Button Switch
- 1 × LED
- 1 × 330Ω Resistor
- 2 × AA Battery Pack
- Jumper Wires

BUILD PROCEDURE

1. Place the push-button switch across the center divider of the breadboard.
2. Wire the switch in series with the resistor and LED.
3. Press the button to complete the loop (closed circuit) and illuminate LED.
4. Release the button to break the connection (open circuit).

Learning: Understand manual switches & current interruption.



Time: 25 mins

Output: Acoustic Sound (Buzzer)

Electronic Quiz Buzzer

> **Objective:** Convert electrical energy into audible sound waves for a fun, competitive classroom game quiz panel.

COMPONENTS USED

- Active Buzzer (Piezo)
- Tactile Push Button
- Battery Supply
- Jumper Wires
- Cardboard Quiz Console

BUILD PROCEDURE

1. Verify the active buzzer's rated polarity (+ and - leads).
2. Connect the buzzer in series with the push button and battery.
3. Press button to produce an instant audible tone.
4. Decorate and install onto a cardboard quiz contestant console.

- Learning: Electrical-to-acoustic energy conversion & signals.



Time: 25 mins

Output: Alarm Sound

Tilt-Activated Alarm

> **Objective:** Engineer an orientation-sensitive security alarm using a mechanical ball-tilt switch sensor.

COMPONENTS USED

- Metallic Tilt Switch
- Active Buzzer
- Mini Breadboard
- Battery Supply
- Jumper Leads

BUILD PROCEDURE

1. Mount the cylindrical tilt switch vertically on the breadboard.
2. Connect in series with the buzzer and battery power source.
3. Observe conductibility as internal metallic ball rolls with gravity.
4. Gently tilt the circuit past 45° to sound the intrusion alarm.

- Learning: Sensor triggering, motion detection, and security circuits.



Time: 30 mins Output: Measured/Visual Light Response

Light Detector

> **Objective:** Discover photo-sensing physics by testing how ambient light changes an LDR's resistance and affects circuit behavior.

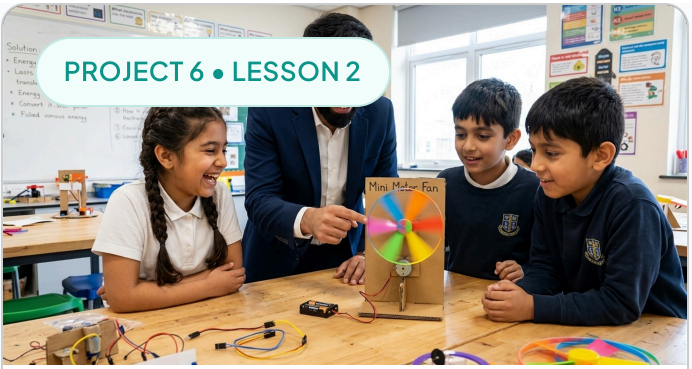
COMPONENTS USED

- Light-Dependent Resistor (LDR)
- 1 × Bright LED
- 1 × 330Ω Resistor
- 2 × AA Batteries
- Mini Breadboard

BUILD PROCEDURE

1. Observe LDR sensor surface under direct room lighting.
2. Build series circuit with LDR, protective resistor, and LED.
3. Cup hands or shade LDR to observe resistance change.
4. Observe how light variations regulate electron current.

Learning: Optical sensors, variable resistance & environmental triggers.



Time: 30 mins Output: Mechanical Rotation & Airflow

Mini Motor Fan

> **Objective:** Convert electrical energy into rotational kinetic motion and aerodynamic airflow using a 3–6V DC motor.

COMPONENTS USED

- Small 3–6V DC Motor
- Lightweight Propeller Blade
- Cardboard Stand
- 2 × AA Battery Pack
- Jumper Leads

BUILD PROCEDURE

1. Inspect motor terminals and verify rated supply voltage.
2. Affix DC motor securely onto a sturdy cardboard support.
3. Press-fit the lightweight plastic propeller fan blade onto shaft.
4. Connect to battery pack and feel the refreshing airflow.

Learning: Electromagnetic rotation, mechanical propulsion & aerodynamics.

CURRICULUM ROADMAP

Structured 4-Hour Schedule

Balanced across 2 structured lessons with hands-on time, student safety briefings, and diagnostic troubleshooting.

LESSON 1

Duration: 2 Hours

Electronics Fundamentals

Theory & Concepts:

- Introduction to electronics and STEM principles
- Battery, voltage, current, and resistance fundamentals
- Conductors vs. insulators
- Breadboard navigation & jumper wire routing
- LED polarity & current-limiting resistor protection
- Open vs. closed circuits & fault testing

Timeline	Workshop Activity
`00-15 min`	Introduction, safety orientation & tool handling
`15-30 min`	Component identification & breadboard exploration
`30-55 min`	**Project 1:** Mini LED Torch construction
`55-80 min`	**Project 2:** Push-Button Light & loop control
`80-105 min`	**Project 3:** Electronic Quiz Buzzer panel
`105-120 min`	Circuit testing, troubleshooting & Q&A assessment

LESSON 2

Duration: 2 Hours

Sensors, Alarms, and Motion

Theory & Concepts:

- Input sensors vs. output transducers
- Tilt switches & gravity-based mechanisms
- Light-dependent resistors (LDRs) and optical variance
- DC electric motors & electro-mechanical conversion
- Miniature propeller aerodynamics & thrust
- Creative STEM design challenges & circuit diagnostics

Timeline	Workshop Activity
`00-15 min`	Review of Lesson 1 & sensor/motor introduction
`15-40 min`	**Project 4:** Tilt-Activated Alarm calibration
`40-70 min`	**Project 5:** Light Detector (LDR) experiments
`70-100 min`	**Project 6:** Mini Motor Fan assembly
`100-115 min`	Creative STEM design challenge & speed testing
`115-120 min`	Comprehensive review, outcome celebration & feedback

HARDWARE BILL OF MATERIALS

Required Kit Components

Every student receives a complete, re-usable hardware kit with safe, solder-free components.

Filter kit components...

14 Standard Kit Items & Guidebook

Component Specification	Educational Purpose	Quantity
Mini Breadboard	Solderless circuit prototyping platform	1
LEDs (Mixed Colours)	Visual output indicators, polarity training	6
Resistors (220Ω, 330Ω, 1kΩ)	Current-limiting and circuit protection	10
2 × AA Battery Holder	Safe, low-voltage power supply enclosure	1
AA Batteries (Alkaline)	Portable DC power source	2
Jumper Wires (Flexible Male-Male)	Rapid solderless circuit connections	10
Push Buttons (Tactile)	Manual switching and signal triggering	2
Active Buzzer	Audible alert and quiz sound signaling	1
Crocodile Clips	Firm mechanical and electrical gripping	2
Storage Box & Labels	Individual kit organization and identification	1
Small DC Motor (3–6V)	Electromechanical kinetic rotation	1
Light-Dependent Resistor (LDR)	Ambient optical environmental sensing	1
Tilt Switch Sensor	Motion and gravitational tilt detection	1

Component Specification	Educational Purpose	Quantity
Detailed Illustrated User Manual & Project Guidebook	Step-by-step schematics, pictorial troubleshooting & bonus home DIY project blueprints	1

Included User Manual & Multi-Project Potential: Every kit contains a comprehensive, professionally illustrated manual. Beyond the 6 core workshop projects, students can follow the manual to build many additional DIY experiments at home (such as Morse code transmitters, multi-LED logic circuits, vibration sensors, and light-sensitive alarms) using the same kit hardware!

Modular Curriculum Expansion: An infrared (IR) obstacle detection sensor module can be seamlessly added later as an optional curriculum extension.

Safety Guidelines

- Use strictly low-voltage safe battery power (3V nominal).
- Always disconnect power before re-routing breadboard connections.
- Always use an appropriate current-limiting resistor with each LED.
- Secure DC motors and fan propellers firmly before applying current.
- Keep hair and loose items clear from spinning propeller blades.
- Continuous adult mentoring provided throughout all activities.

Expected Outcomes

- Draw and explain basic schematic circuit loops and polarities.

- Explain why current-limiting resistors prevent LED burnout.
- Demonstrate manual switching on both series and open configurations.
- Calibrate environmental response using tilt and light sensor circuits.
- Independently detect and resolve at least one real circuit fault.

Project	Primary STEM Concept	Tangible Output
1. Mini LED Torch	Complete Circuit & LED Polarity	Controlled Illumination
2. Push-Button Light	Mechanical Switching & Interruption	Interactive Light Toggle
3. Electronic Quiz Buzzer	Electrical-to-Sound Energy Conversion	Audible Game Tone
4. Tilt-Activated Alarm	Gravitational Orientation Sensing	Motion-Activated Siren
5. Light Detector	Light-Sensitive Resistance (LDR)	Optical Sensor Modulation
6. Mini Motor Fan	Electrical-to-Mechanical Energy Conversion	High-Speed Propeller Airflow